Adapting Pretrained Language Models for Citation Classification via Self-Supervised Contrastive Learning

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KDD2025 TORONTO

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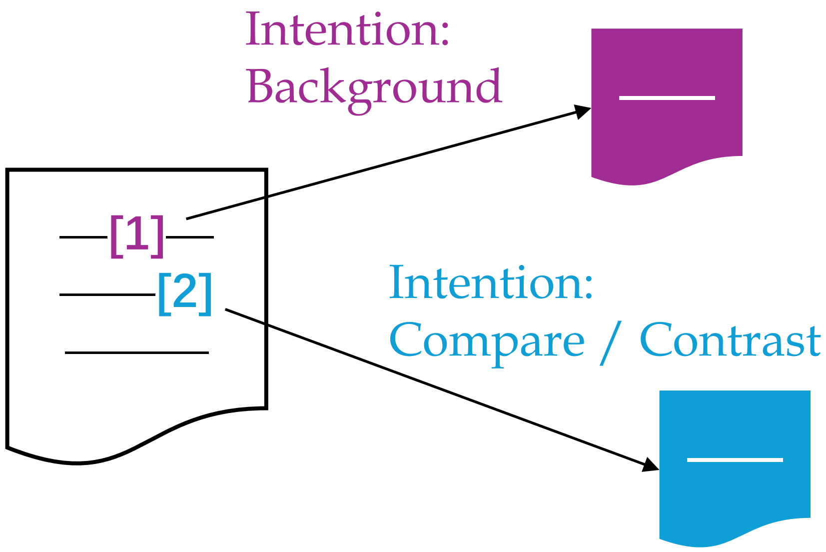
Introduction

Citation Classification: why authors cite other papers?



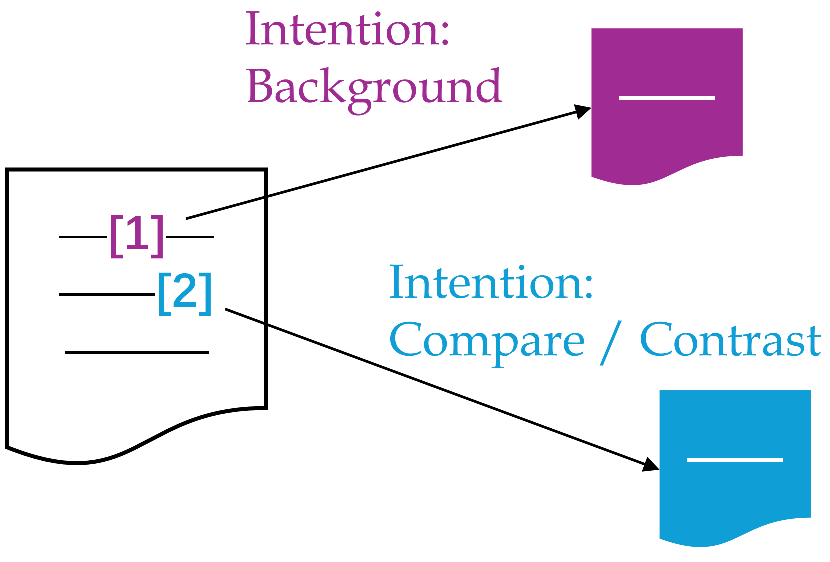
Introduction

Citation Classification: why authors cite other papers?

The citation context:

.....As computer vision techniques have improved over the past decade, other research has begun directly using visual information in place of feature norms. The first work to do this with topic models is [1]. They use a Bag of Visual Words model to create a bimodal vocabulary describing documents. The topic model using the bimodal vocabulary outperforms a purely textual based model in word association and word similarity prediction. Bruni et al. [2] show how a BoVW model may be easily combined with a distributional vector space model of language....

citation anchor



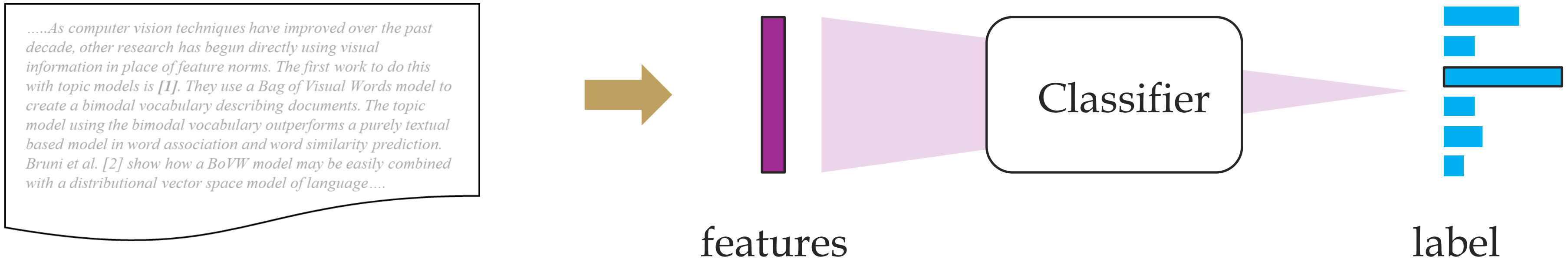
Related Works

Hand-engineered features:

- Cue words, metadiscourse, part-of-speech tags, dependency relationships…

Deep-learned features:

- Word representation from popular embedding models (Glove, ElMo, etc.)



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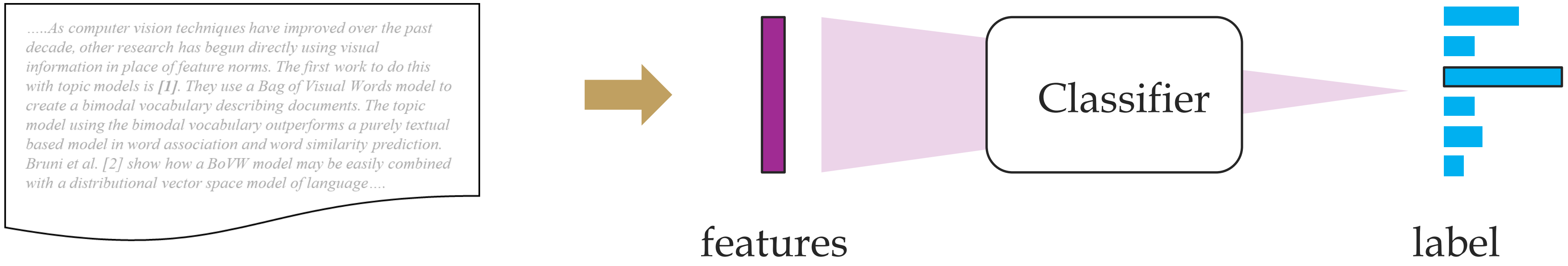
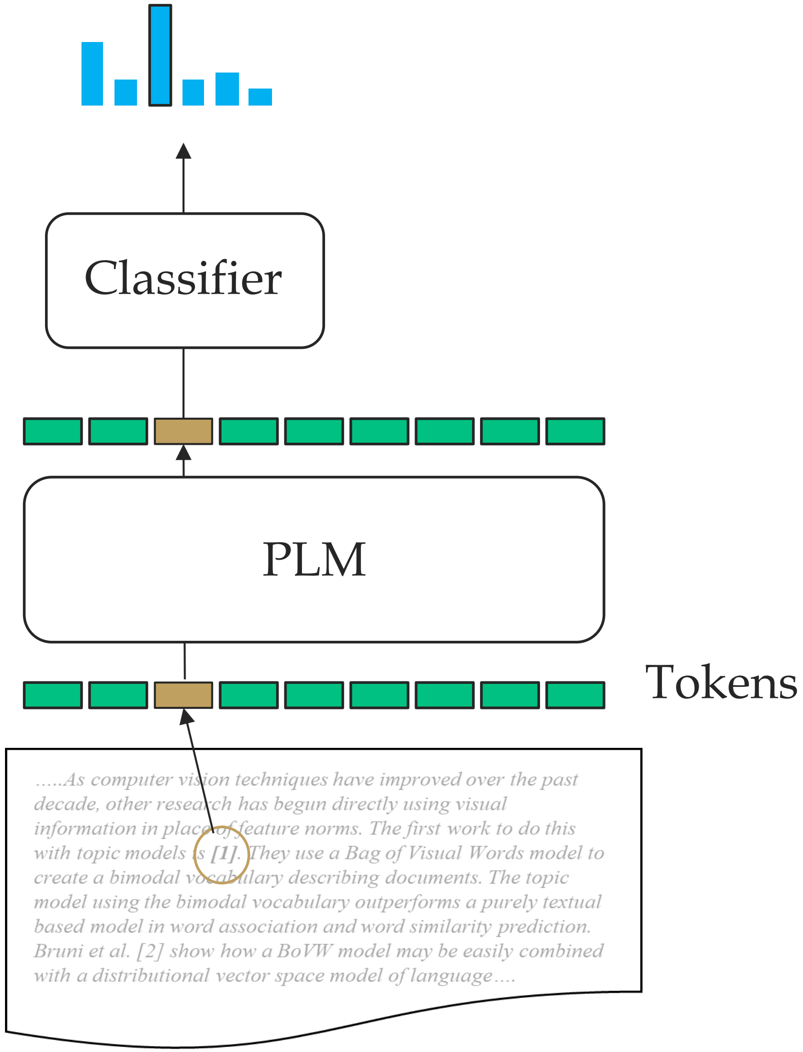


These methods struggle to capture the semantic nuances between different intentions due to limited model capacity.

Opportunities

PLMs have acquired extensive linguistic knowledge during pretraining.

PLMs can be fine-tuned on citation classification datasets to discern subtle semantic nuances in citation intentions.



Opportunities

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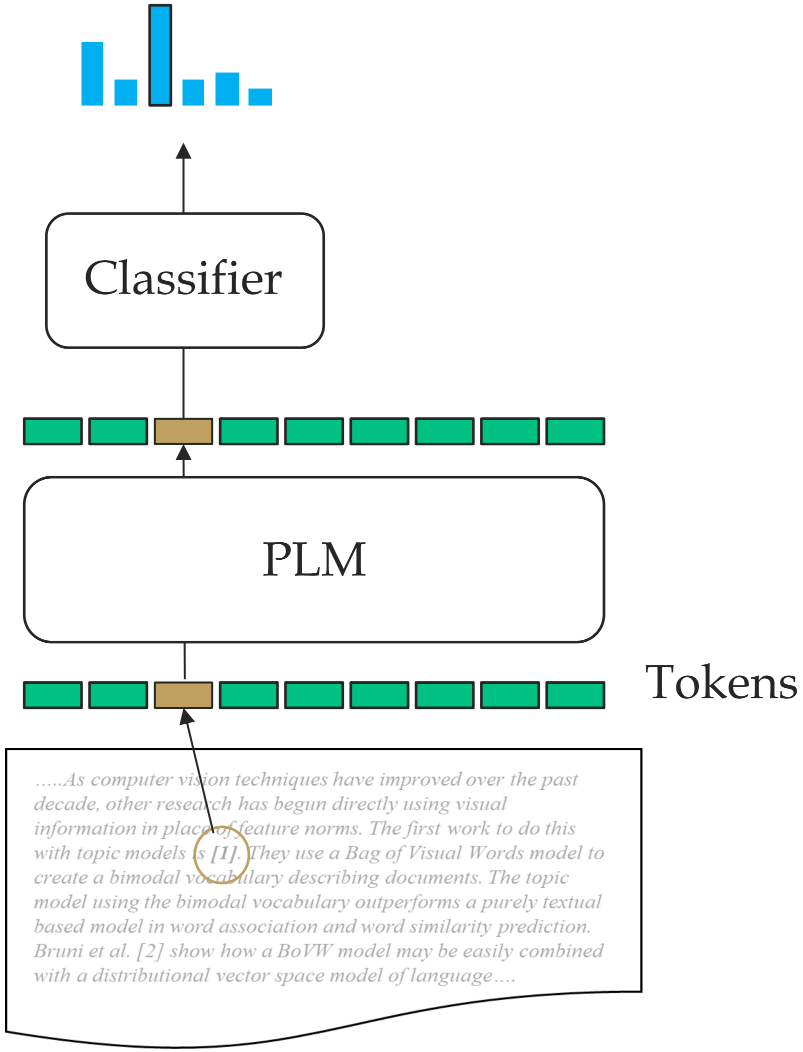
However, there are a few unique challenges for citation classification.



Challenges

1. **Scarcity of labeled citation data.**

- Expensive data collection:

  Author self-annotation, employ annotators with academic backgrounds.

- Collected datasets fall short in manifesting the task-specific patterns and restrain the performance of PLMs.

Challenges

2. **Defocusing on the target citation.**

*The first work to do this with topic models is **[1]**.*

Traditional methods:
- "Citance": the citing sentence
- Highly-related context.
- Easy for feature design
- No long-range dependencies

Challenges

2. Defocusing on the target citation.

*The first work to do this with topic models is **[1]**.*

target citation

*.....As computer vision techniques have improved over the past decade, other research has begun directly using visual information in place of feature norms. The first work to do this with topic models is **[1].** They use a Bag of Visual Words model to create a bimodal vocabulary describing documents. The topic model using the bimodal vocabulary outperforms a purely textual based model in word association and word similarity prediction. **Bruni et al. [2]** show how a BoVW model may be easily combined with a distributional vector space model of language....*

other citations

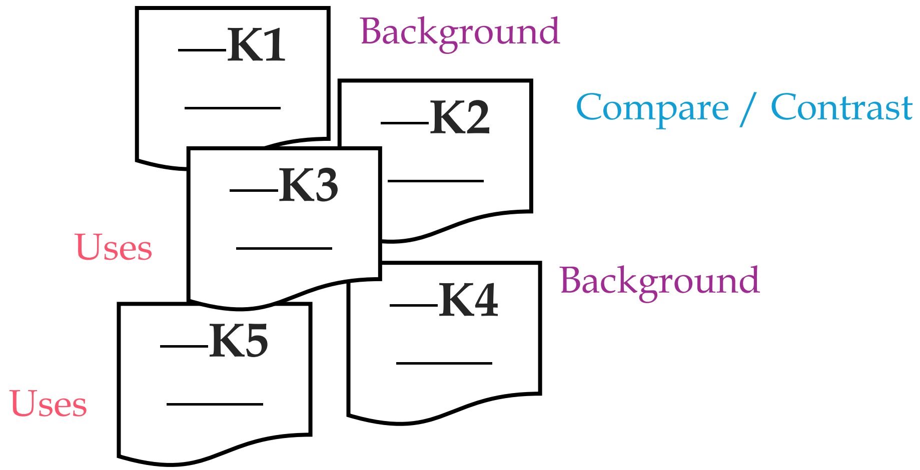
Traditional methods:
- "Citance": the citing sentence
- Highly-related context.
- Easy for feature design
- No long-range dependencies

With PLMs:
- Longer input for more information
- But excessive noise.
- Defocusing.

Challenges

3. Spurious correlation based on keyphrases

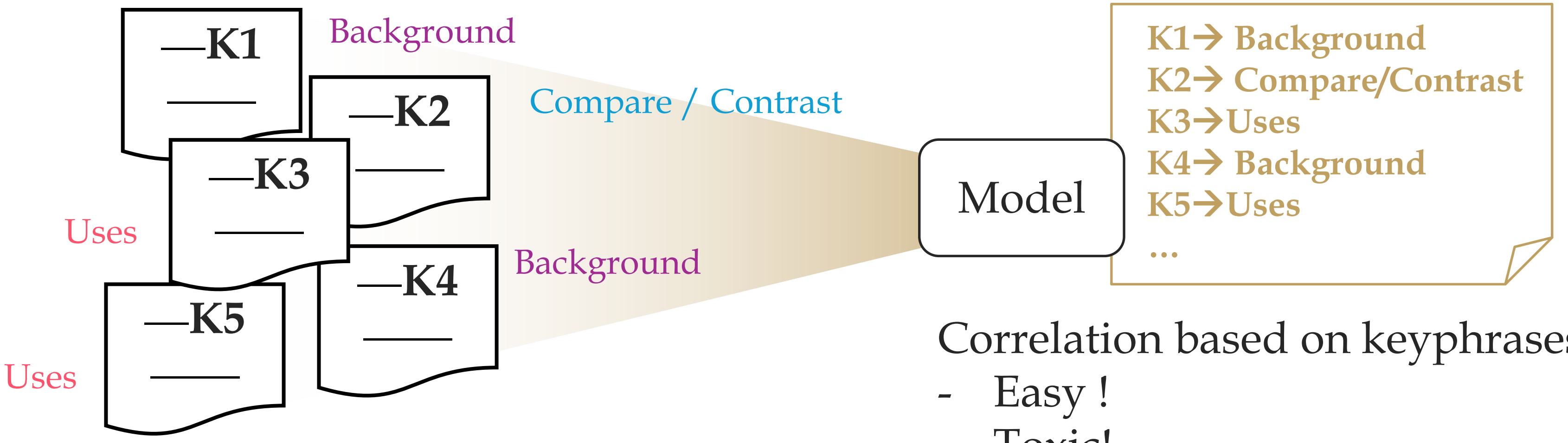
- Scientific keyphrases (STKs): the name of research subjects, tasks, and techniques…
- STKs are repeatedly mentioned and semantically significant during pretraining.



Challenges

3. Spurious correlation based on keyphrases

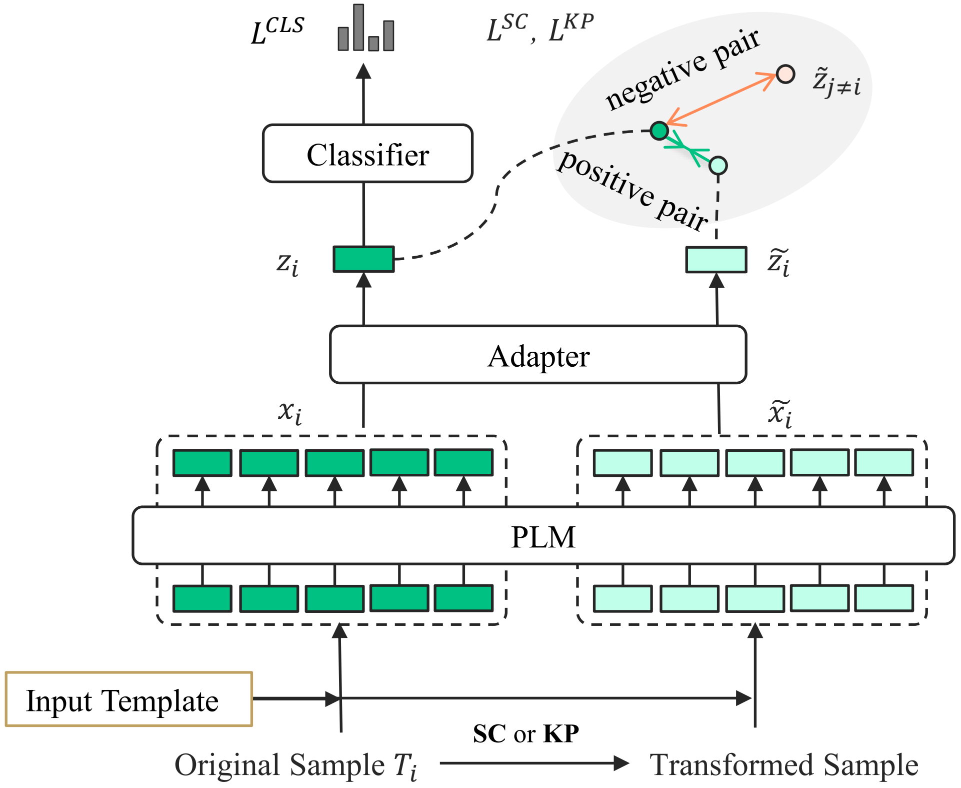
- Scientific keyphrases (STKs): the name of research subjects, tasks, and techniques…
- STKs are repeatedly mentioned and semantically significant during pretraining.
- But not indicative for citation intention label.



Correlation based on keyphrases
- Easy !
- Toxic!

Overview

Our framework adapts PLMs for Citation classification via self-supervised contrastive learning.



Citss:

Generate contrastive pairs in a self-supervised manner and derive contrastive loss to provide extra supervision signals to address Challenge 1.

Two transform strategies.

Sentence-level cropping for Challenge 2.

Keyphrase perturbation for Challenge 3.

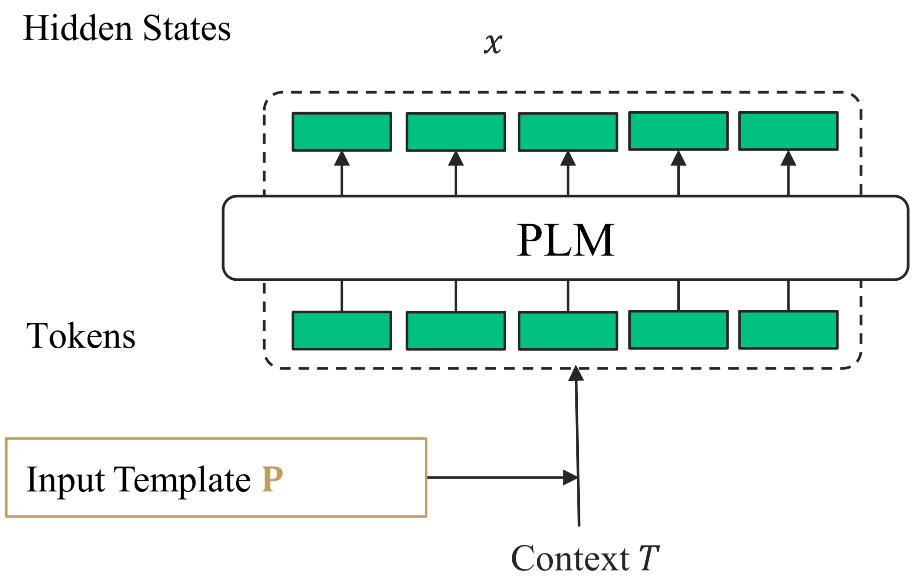
Architecture

Interaction with PLMs

$$x = \mathcal{M}(T, \mathbf{P})$$

Encoder-based PLMs: Readout at [MSK]

Decoder-based LLMs: "Explicit-One-word-Limitation"



Architecture

Interaction with PLMs

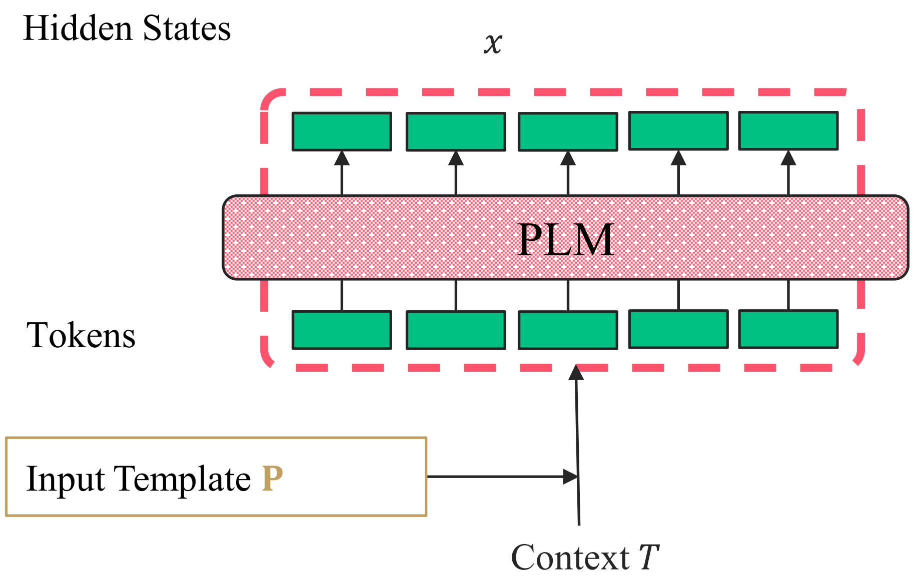
$$x = \mathcal{M}(T, \mathbf{P})$$

Encoder-based PLMs: Readout at [MSK]

Decoder-based LLMs: "Explicit-One-word-Limitation"

Compatible with **Parameter Efficient Fine-Tuning** techniques.

- LoRA, etc.
- Further reduce the trainable parameter.



Architecture

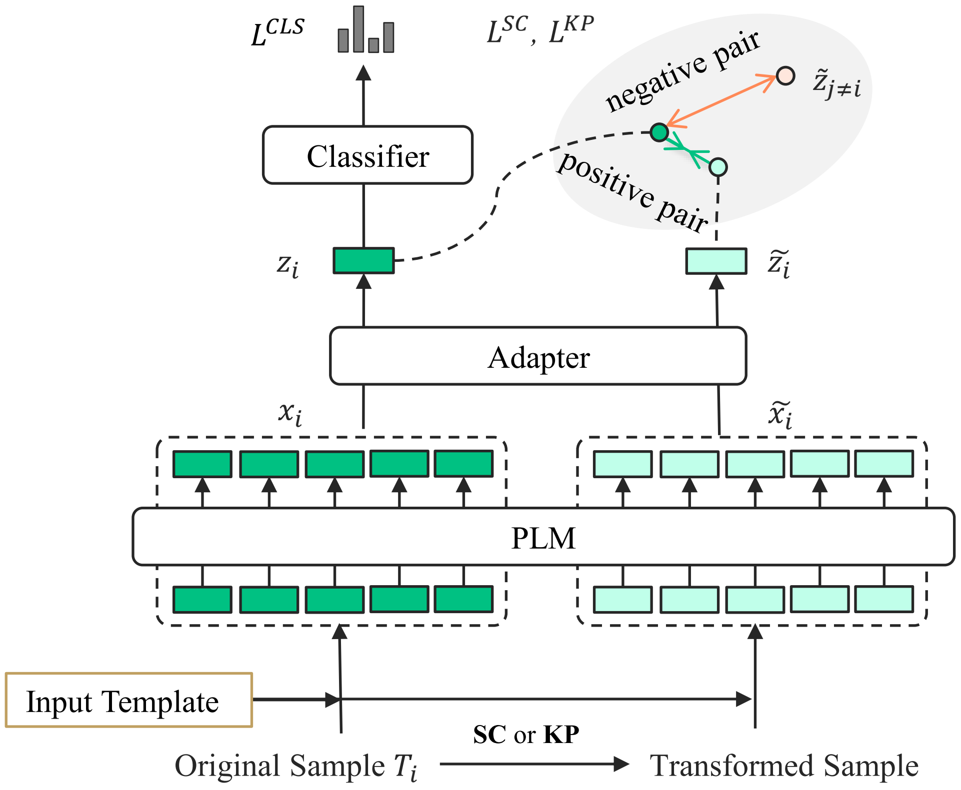
MLP Adapter $$f(x) = W_2 \mathrm{LN}(\mathrm{GeLU}(W_1 x + b_1)) + b_2$$

Classifier $$y_i = g(z_i) = \mathrm{softmax}(W_3 z_i + b_3),$$

Training Loss $$L = L^{CLS} + \lambda_1 L^{SC} + \lambda_2 L^{KP} + \omega L^{pnt}$$

$$L^{CLS} = \sum_{i \in \mathcal{B}} CrossEntropy(y_i, \hat{y}_i),$$

$$L^{InfoNCE} = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \log \frac{\exp sim(z_i, \tilde{z}_i)}{\sum_{j \in \mathcal{B}} \exp sim(z_i, \tilde{z}_j)}$$

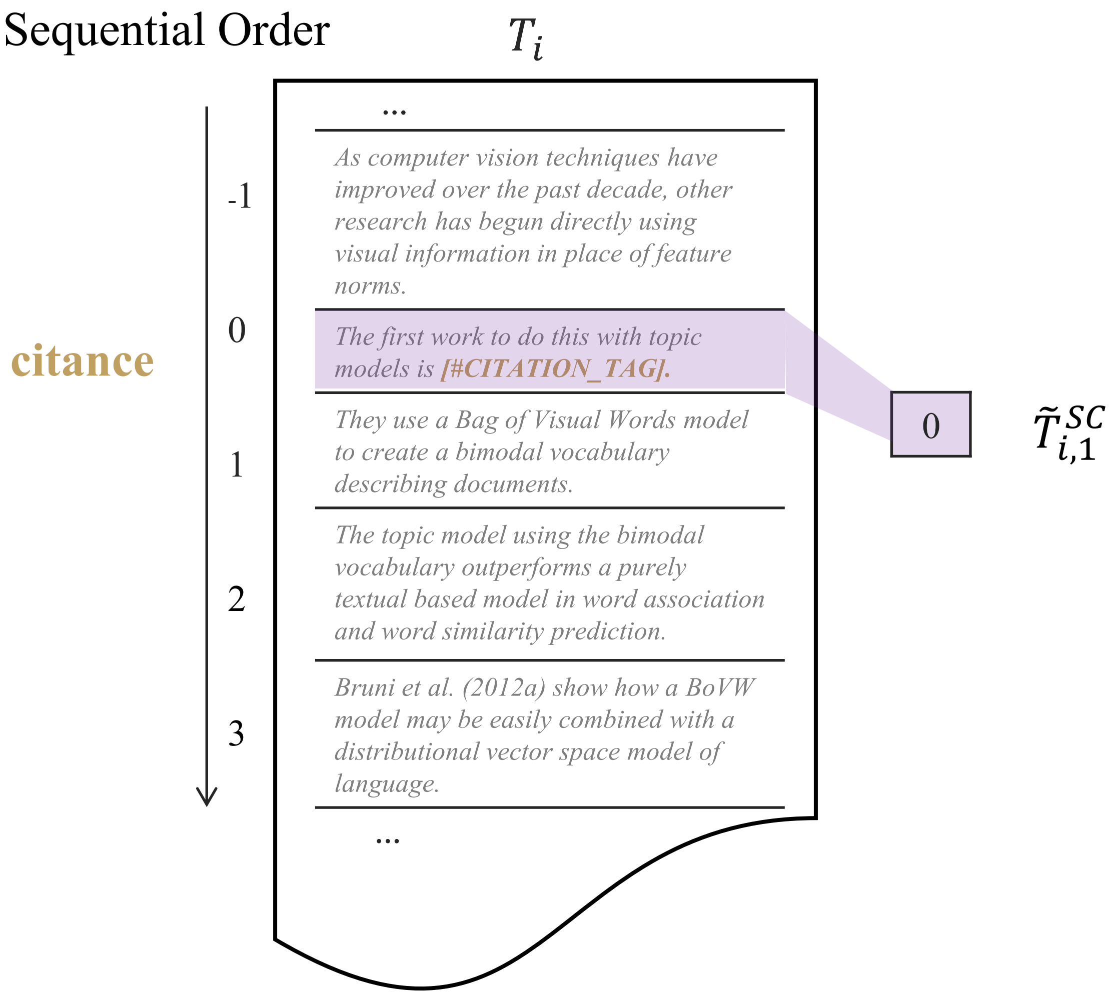


Sentence-Level Cropping

Randomly crops a subsequence with the same citance but different context ranges.

$$T_i = \langle s_i^{-l}, \ldots, s_i^{-1}, s_i^{0}, s_i^{1}, \ldots, s_i^{l} \rangle$$

$$\mathcal{SC}(T_i) = \{\langle s_i^{-b}, \ldots, s_i^{0}, \ldots, s_i^{v} \rangle \mid \forall b, v, -l \leq -b \leq 0 \leq v \leq l\} / \{T_i\}$$



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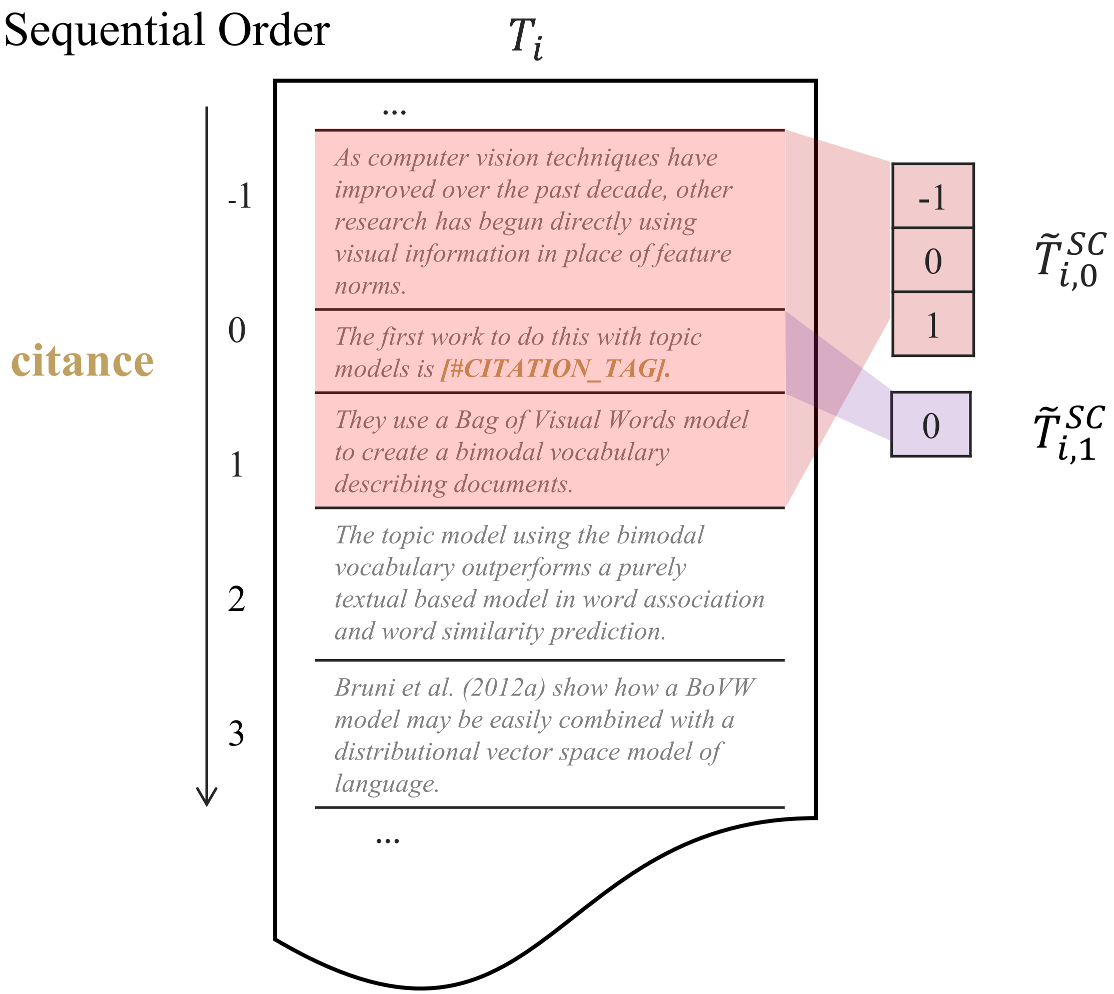

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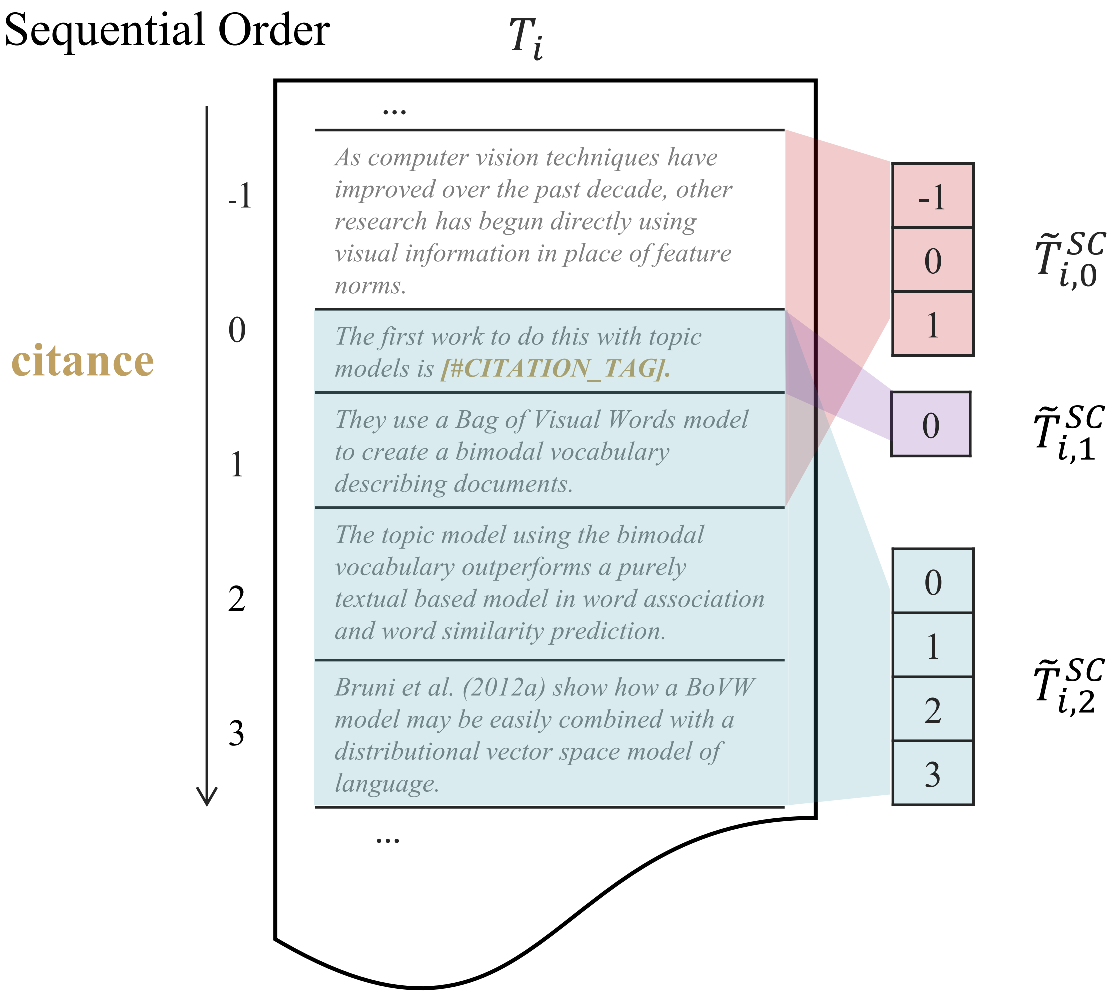

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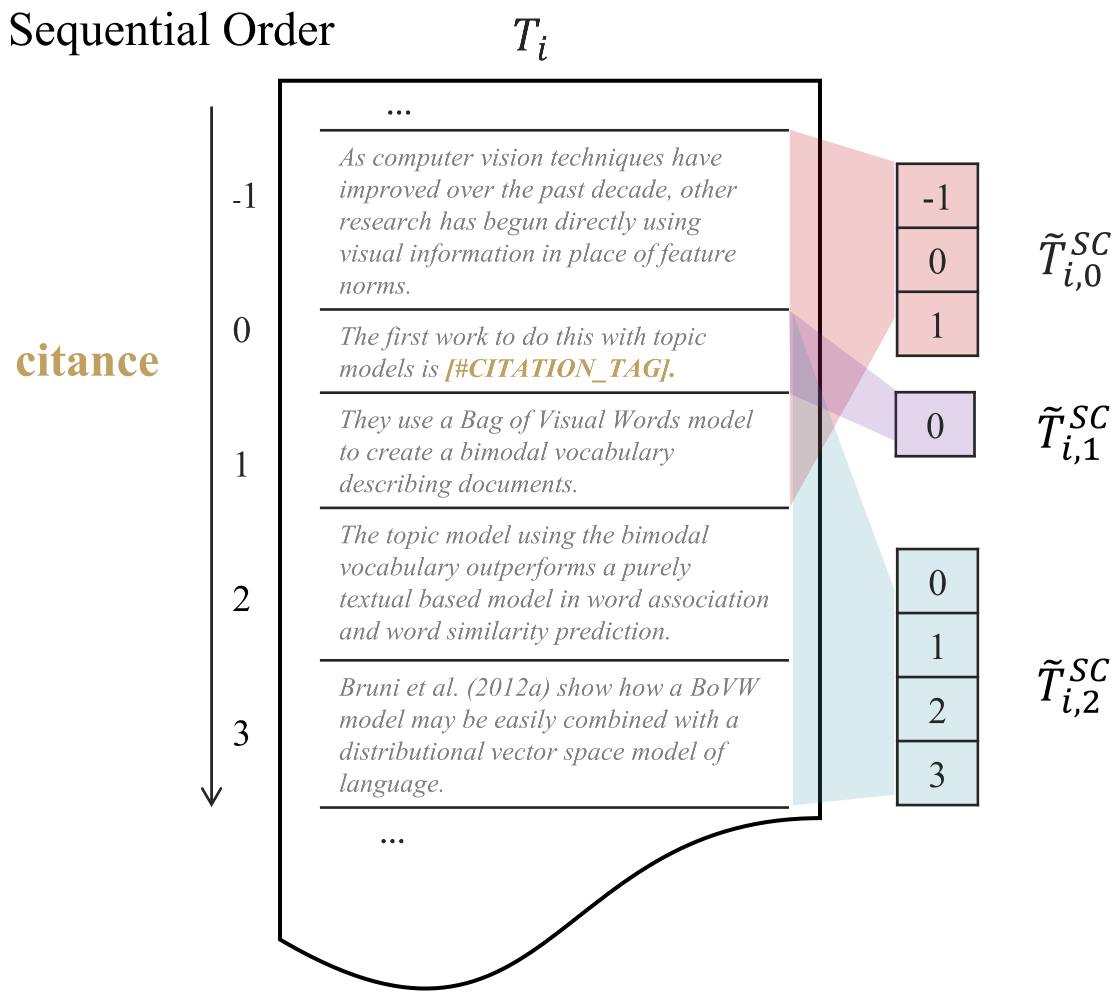
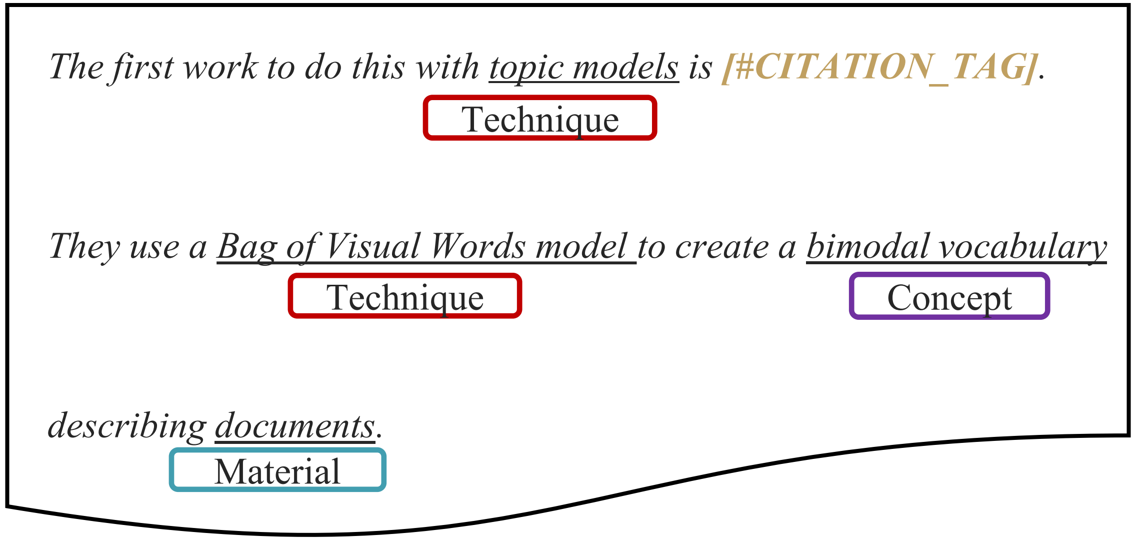
Intuition: "image cropping"

Attend to the target object against background noise.



Keyphrase Perturbation

Intuition: Modifying the STKs usually does not affect the citation intention.



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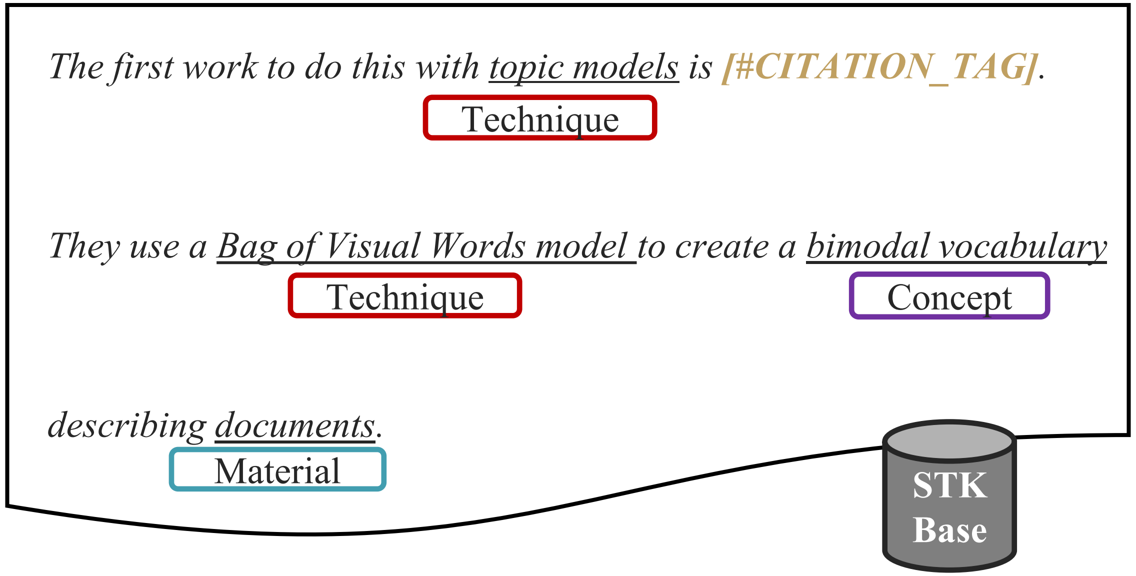
Use LLMs to extract STKs and form a STK Base.



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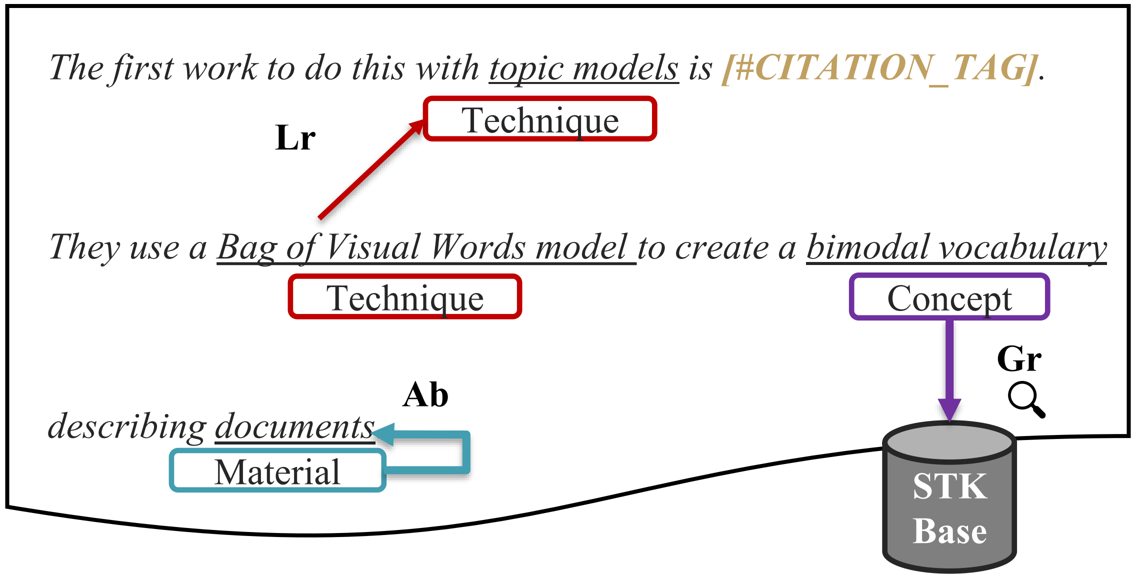
Three operations for STKs

- **G**lobal **r**eplacement
- **L**ocal **r**eplacement
- **Ab**straction



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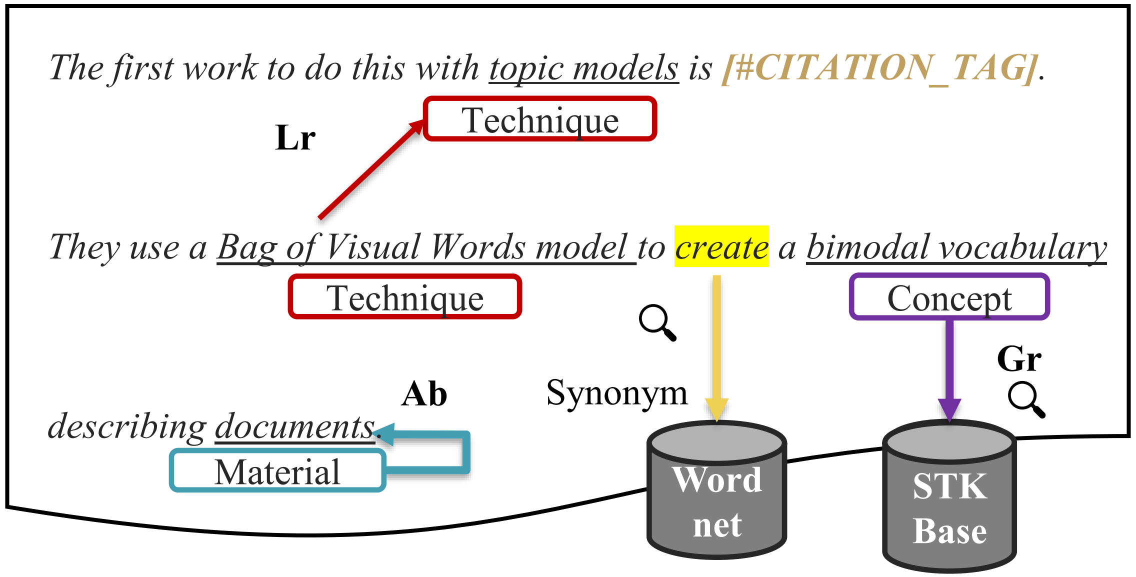
Random synonym replacement.



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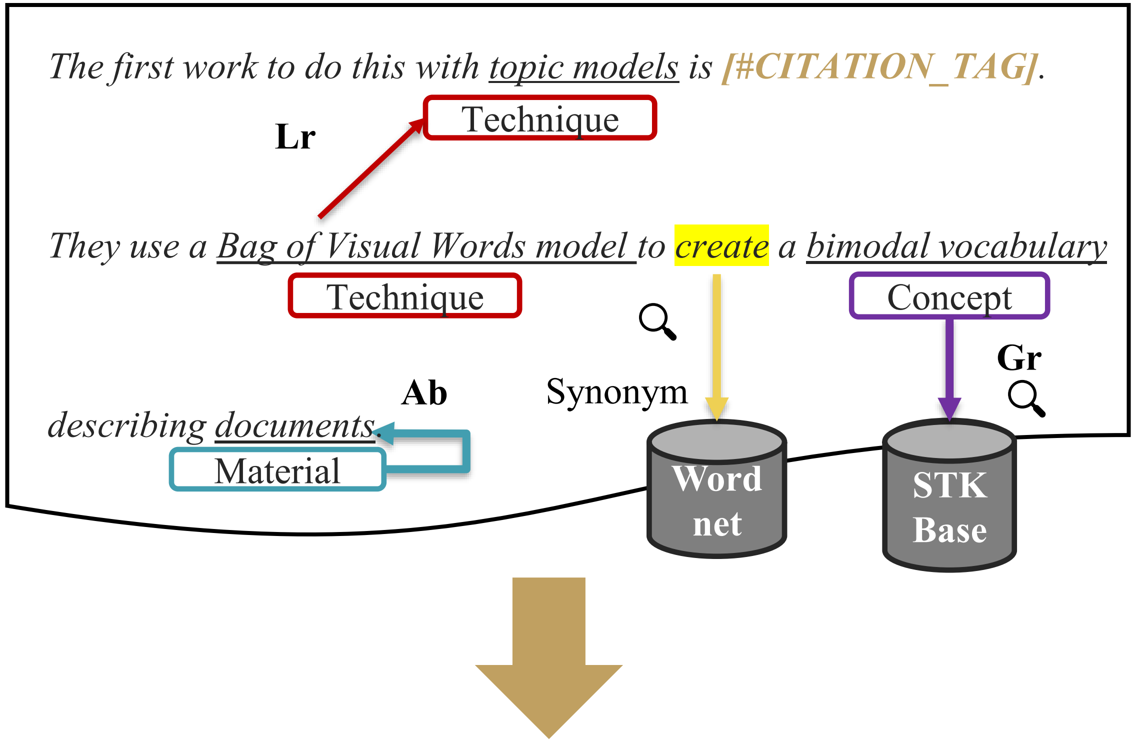

The first work to do this with Bag of Visual Words model is ***[#CITATION_TAG]****. They use a Bag of Visual Words model to build a semantic representation describing Material-1.*

Overall Performance

Compare with fine-tuning and in-context learning baselines,

1. Consistent superiority of **Citss** with 2 backbones (SciBERT, Llama3-8B) on 3 datasets.

2. **Citss** can effectively fine-tune LLMs with extremely limited citation data.

Method	Backbone	ACL-ARC Macro-f1	ACL-ARC Accuracy	FOCAL Macro-f1	FOCAL Accuracy	ACT2 Macro-f1	ACT2 Accuracy
Scaffold	NA	0.496 ± 0.021	0.649 ± 0.012	0.145 ± 0.041	0.447 ± 0.133	0.146±0.006	0.363±0.017
IREL	SciBERT	0.614 ± 0.037	0.721 ± 0.025	0.580 ± 0.086	0.742 ± 0.007	**0.262**±0.012	0.468±0.028
TRL	SciBERT	0.476 ± 0.024	0.610 ± 0.007	0.604 ± 0.009	<u>0.756</u> ± 0.009	0.118±0.001	<u>0.544</u>±0.000
PET	SciBERT	0.616 ± 0.022	0.723 ± 0.019	0.641 ± 0.044	0.750 ± 0.008	0.258± 0.018	0.537±0.020
Citss	SciBERT	**0.665*** ± 0.018	**0.743** ± 0.006	**0.679** ± 0.023	**0.777*** ± 0.005	0.254±0.012	**0.563***±0.009
IFP	Llama3-8B	0.422	0.575	0.243	0.398	0.213	0.446
LoRA	Llama3-8B-bfloat16	0.670 ± 0.050	0.745 ± 0.028	0.670 ± 0.035	0.757±0.001	0.242 ± 0.034	0.529 ± 0.014
Citss + LoRA	Llama3-8B-bfloat16	**0.744** ± 0.010	**0.819*** ± 0.007	**0.682** ± 0.024	**0.768*** ± 0.001	**0.266** ± 0.006	**0.549** ± 0.017
IFP	Llama3-70B	0.569	0.701	0.430	0.623	0.242	0.545

Ablation Study

Each of the transform strategy improve the performance.

Combing SC and KP further improves.

SciBERT	ACL-ARC Macro-f1	ACL-ARC Accuracy	Imp.(%)	FOCAL Macro-f1	FOCAL Accuracy	Imp.(%)	ACT2 Macro-f1	ACT2 Accuracy	Imp.(%)
$\lambda_1, \lambda_2 = 0$	0.616 ± 0.012	0.714 ± 0.022	-	0.641 ± 0.029	0.742 ± 0.004	-	0.262 ± 0.016	0.539 ± 0.021	-
$\lambda_2 = 0$	0.660 ± 0.025	0.745 ± 0.032	5.7	0.666 ± 0.029	0.762 ± 0.006	3.3	0.246 ± 0.036	0.558 ± 0.008	0.3
$\lambda_1 = 0$	0.646 ± 0.032	0.730 ± 0.017	3.5	0.654 ± 0.004	0.764 ± 0.009	2.5	0.260 ± 0.003	0.549 ± 0.017	0.9
Llama3-8B	**Macro-f1**	**Accuracy**	**Imp.(%)**	**Macro-f1**	**Accuracy**	**Imp.(%)**	**Macro-f1**	**Accuracy**	**Imp.(%)**
$\lambda_1, \lambda_2 = 0$	0.729 ± 0.007	0.799 ± 0.003	-	0.673 ± 0.008	0.762 ± 0.003	-	0.227 ± 0.006	0.544 ± 0.017	-
$\lambda_2 = 0$	0.736 ± 0.021	0.805 ± 0.013	0.9	0.681 ± 0.009	0.769 ± 0.004	1.0	0.248 ± 0.016	0.542 ± 0.016	2.5
$\lambda_1 = 0$	0.739 ± 0.008	0.804 ± 0.011	1.0	0.675 ± 0.005	0.766 ± 0.004	0.4	0.236 ± 0.009	0.556 ± 0.019	2.8

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Check our paper for more results!

Thanks!

Code: github.com/LITONG99/Citss

Contact: tlice@connect.ust.hk

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